Т5) $\xi \sim R[\theta; 2\theta]$

а) $\vec{x}_n$ – выборка

$$M\xi = \int_{-\infty}^{\infty} x\,dF = \int_{\theta}^{2\theta} x \cdot \frac{1}{\theta}\,dx = \frac{1}{\theta} \cdot \frac{x^2}{2}\Big|_{\theta}^{2\theta} = \frac{1}{\theta}\left(2\theta^2 - \frac{\theta^2}{2}\right) =$$

$$= \frac{3}{2}\theta$$

$$M\xi^2 = \int_{-\infty}^{\infty} x^2\,dF = \int_{\theta}^{2\theta} x^2 \cdot \frac{1}{\theta}\,dx = \frac{1}{\theta}\left(\frac{8\theta^3}{3} - \frac{\theta^3}{3}\right) =$$

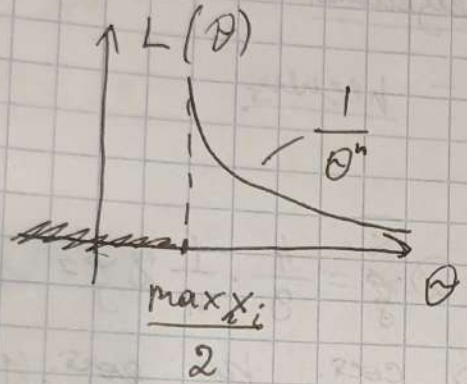
$$= \frac{7}{3}\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = \frac{7}{3}\theta^2 - \frac{9}{4}\theta^2 = \frac{1}{12}\theta^2$$

$\alpha_1(\theta) = \tilde{\alpha}_1$ $\qquad$ $\lambda^* - \hat{h} < \Lambda_1 \ldots) \approx \beta$

$\frac{3}{2}\theta = \bar{X} \to \tilde{\theta}_1 = \frac{2}{3}\bar{X}$ $\qquad$ ОММ

$$L(\theta) = \prod_{i=1}^{n} p(x,\theta) = \frac{1}{\theta^n}\{(\theta \le x_i \le 2\theta)\}$$



$\cancel{\theta \le \max x_i \le 2\theta}$

$\cancel{\sup_{\theta>0} L(\theta) \to \tilde{\theta} =}$

$\min x_i \ge \theta; \quad \max x_i \le 2\theta$

$\min x_i \le \theta \le \frac{\max x_i}{2}$

$\sup_{\theta>0} L(\theta) \to \tilde{\theta}_2 = \frac{\max x_i}{2} = \frac{X_{max}}{2}$

ОМП

б) несмещ.: $M\tilde{\theta}_1 = \frac{2}{3} M\bar{X} = \frac{2}{3}\cdot \frac{1}{n}\cdot n M\xi =$

$= \frac{2}{3}\cdot\frac{3}{2}\theta = \theta$ — несмещенная

$M\tilde{\theta}_2 = M\frac{X_{max}}{2}$

$X_{max} \sim \underbrace{(F(x))^n}_{\Psi(x)}$ $\qquad$ $\psi(x) = \Psi'(x) = n\left(\frac{x-\theta}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}\cdot$

$\cdot\{(\theta; 2\theta)\}$

$$M X_{max} = \int_{\theta}^{2\theta} x\cdot n\left(\frac{x-\theta}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}dx =$$

$$\cancel{= \frac{n}{\theta^n}\cdot\frac{(2\theta)^{n+1}}{n+1}\cdot\theta^{n+1} = \frac{n}{n+1}\cdot 2^{n+1}\cdot\theta^{n+1}\cdot\theta}$$

$$= \left\{\frac{x-\theta}{\theta} = t;\ x = \theta t + \theta;\ dx = \theta dt\right\} = \int_0^1 (\theta t + \theta)\frac{n\theta}{\theta^n} t^{n-1} dt =$$

$$= \int_0^1 n(\theta t^n + \theta t^{n-1})\,dt = n\theta\left(\frac{t^{n+1}}{n+1} + \frac{t^n}{n}\right)\Big|_0^1 =$$

$$= n\theta\cdot\left(\frac{1}{n+1}+\frac{1}{n}\right) = \cancel{n}\theta\cdot\frac{n+n+1}{(n+1)\cancel{n}} = \theta\cdot\frac{2n+1}{n+1}$$

$$M\frac{X_{max}}{2} = \frac{2n+1}{2n+2}\theta \neq \theta$$ – <u>смещенная</u>

$$\tilde{\theta}_2' = \frac{2n+2}{2n+1}\,\frac{X_{max}}{2} = \frac{n+1}{2n+1}X_{max}$$ – <u>несмещ.</u>

<u>состоятельность:</u>

$$D\tilde{\theta}_1 = \cancel{\frac{1}{12}}\,D\left(\frac{2}{3}\bar{X}\right) = \frac{4}{9}D\bar{X} = \frac{4}{9}\cdot\frac{1}{n}D\xi =$$

$$= \frac{4}{9}\cdot\frac{1}{n}\left(\frac{1}{12}\theta^2\right)\xrightarrow[n\to\infty]{} 0 \Rightarrow$$ <u>сост.</u> по дост. усл.

$$D\tilde{\theta}_2' = D\left(\frac{n+1}{2n+1}X_{max}\right) = \left(\frac{n+1}{2n+1}\right)^2 DX_{max}$$

$$MX_{max}^2 = \int_\theta^{2\theta} x^2\cdot n\left(\frac{x-\theta}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}\,dx = \left\{\frac{x-\theta}{\theta} = t\right\} =$$

$$= \int_0^1 (\theta t+\theta)^2\cdot n\cdot t^{n-1}\cdot\frac{\theta}{\theta}\,dt = n\theta^2\int_0^1 (t^{n+1}+2t^n+t^{n-1})\,dt =$$

$$= n\theta^2\left(\frac{1}{n+2}+2\cdot\frac{1}{n+1}+\frac{1}{n}\right) = \frac{\theta^2(n^2+n+2n^2+4n+n^2+3n+2)}{(n+2)(n+1)} =$$

$$= \frac{\theta^2(4n^2+8n+2)}{(n+2)(n+1)} = \theta^2\cdot 2\cdot\frac{2n^2+4n+1}{n^2+3n+2} = \cancel{2\theta\cdot\frac{2+\frac{4}{n}+\frac{1}{n^2}}{1+\frac{3}{n}+\frac{2}{n^2}}}$$

$$\cancel{\xrightarrow[n\to\infty]{} 4\theta}$$

$$DX_{max} = 2\theta^2\cdot\frac{2n^2+4n+1}{n^2+3n+2} - \theta^2\cdot\left(\frac{2n+1}{n+1}\right)^2 = \cancel{\frac{4n^2+8n+2-2n^2-3n-1}{(n+1)(n+2)}\theta^2} =$$

$\beta \cdot 5000 \quad \frac{\alpha}{2} \cdot 5000$

$\cancel{= 2n^2+5n+1}$

$$\cancel{\theta^2 \cdot \frac{4n^2+8n+2-2n^2-4n-n-2}{(n+1)(n+2)} = \theta^2 \cdot \frac{2n^2+3n}{(n+1)(n+2)}}$$

$$\cancel{D\tilde{\theta}_2' = \frac{(n+1)n(2n+3)}{(2n+1)^2(n+2)} = \frac{(n^2+n)(2n+3)}{(n+2)(4n^2+4n+1)} = \frac{2n^3+3n^2+2n^2+3n}{4n^3+4n^2+n+8n^2+8n+2} =}$$

$$\cancel{= \frac{2+\frac{3}{n}+\frac{2}{n}+\frac{3}{n^2}}{4+}}$$

$$\Leftrightarrow \theta^2 \cdot \frac{(4n^2+8n+2)(n+1)-(4n^2+4n+1)(n+2)}{(n+1)^2(n+2)} =$$

$$= \theta^2 \cdot \frac{\cancel{4n^3}+\cancel{8n^2}+2n+\cancel{4n^2}+\cancel{8n}+\cancel{2}-\cancel{4n^3}-\cancel{4n^2}-\cancel{n}-\cancel{8n^2}-\cancel{8n}-\cancel{2}}{(n+1)^2(n+2)} =$$

$$= \theta^2 \cdot \cancel{2n} \quad \frac{n}{(n+2)(n+1)^2}$$

$$D\tilde{\theta}_2' = \theta^2 \cdot \frac{n}{(2n+1)^2(n+2)} = \theta^2 \cdot \frac{n}{(n+2)(4n^2+4n+1)} =$$

$$= \theta^2 \cdot \frac{n}{4n^3+4n^2+n+8n^2+8n+2} = \theta^2 \cdot \frac{n}{4n^3+12n^2+9n+2} =$$

$$= \theta^2 \cdot \frac{1}{4n^2+12n+9n+\frac{2}{n}} \xrightarrow[n\to\infty]{} 0. \Rightarrow \underline{\text{сход.}} \text{ по дост. усл.}$$

c) $D\tilde{\theta}_1 \geq D\tilde{\theta}_2'$

$$\frac{\cancel{\theta_3^2}}{27n} > \frac{n\theta^2}{(2n+1)^2(n+2)}$$

$(2n+1)^2(n+2) > 27n^2 \Rightarrow \tilde{\theta}_2'$ огр. $\tilde{\theta}_1$

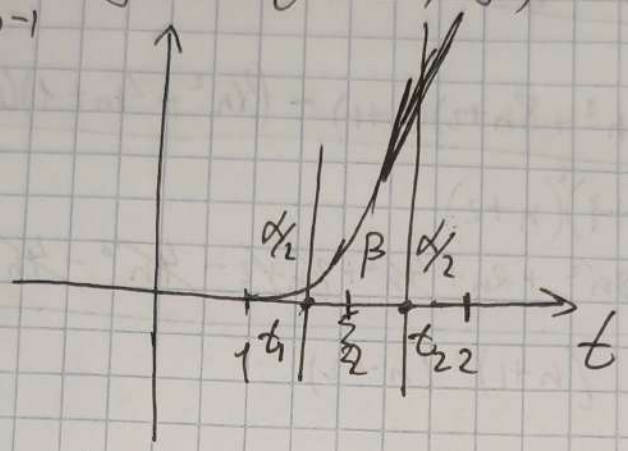
~~[illegible]~~

d) $f(\vec{x}_n, \theta) = \frac{x_{max}}{\theta} \sim n(t-1)^{n-1}$ $t \in [1;2]$

$F_t(t) = P\left(\frac{x_{max}}{\theta} \le t\right) = P(x_{max} \le t\theta) = \left(\frac{t\theta - \theta}{\theta}\right)^n =$

$= (t-1)^n$ (для $t \in [1;2]$)

$\rho_t = n(t-1)^{n-1}$



$t_1 = \rho_{\frac{1-\beta}{2}}$ $\quad t_2 = \rho_{\frac{1+\beta}{2}}$

$F(t) = (t_1 - 1)^n = \frac{1-\beta}{2}$ $\quad (t_2-1)^n = \frac{1+\beta}{2}$

$t_1 = 1 + \left(\frac{1-\beta}{2}\right)^{1/n}$ $\quad t_2 = 1 + \left(\frac{1+\beta}{2}\right)^{1/n}$

$P\left(t_1 < \frac{x_{max}}{\theta} < t_2\right) = \beta$

$\frac{x_{max}}{t_2} < \theta < \frac{x_{max}}{t_1}$

e) <u>ОММ.</u> $\tilde{\theta}_1 = g(\tilde{\alpha}) = \frac{2}{3}\bar{x} = \frac{2}{3}\tilde{\alpha}_1$

$\sqrt{n}\,\frac{g(\tilde{\alpha}) - g(\alpha)}{\sigma(\tilde{\alpha})} \to N(0;1)$

$\sigma^2(\alpha) = \nabla g^T(\alpha)\, K\, \nabla g(\alpha)$

$\frac{\alpha}{2}\cdot 5000$ $\quad \beta\cdot 5000$ $\quad \frac{\alpha}{2}\cdot 5000$

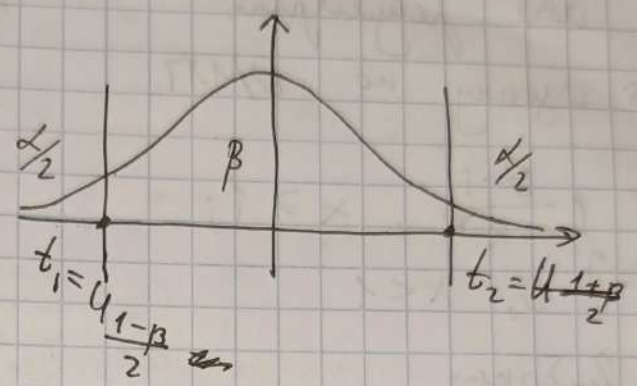
$$\nabla g(\alpha) = \frac{2}{3}$$

$$K_{11} = \alpha_{1+1} - \alpha_1 \alpha_1$$

$$\sigma^2(\alpha) = \frac{2}{3}\left(\alpha_2 - \alpha_1^2\right)\cdot\frac{2}{3} = \frac{4}{9}\left(\alpha_2 - \alpha_1^2\right)$$

$$\sqrt{n}\cdot\frac{\tilde{\theta}_1 - \theta}{\frac{2}{3}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} \rightsquigarrow N(0;1)$$

$$\tilde{\alpha}_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2$$



$$u_{\frac{1-\beta}{2}} < \sqrt{n}\,\frac{\tilde{\theta}_1 - \theta}{\frac{2}{3}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} < u_{\frac{1+\beta}{2}}$$

$$t_1 < \frac{3}{2}\sqrt{n}\,\frac{\tilde{\theta}_1 - \theta}{\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} < t_2$$

$$\frac{2}{3\sqrt{n}}\,t_1\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2} < \tilde{\theta}_1 - \theta < \frac{2}{3\sqrt{n}}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}\,t_2$$

$$\tilde{\theta}_1 - \frac{2}{3\sqrt{n}}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}\,t_2 < \theta < \tilde{\theta}_1 - \frac{2}{3\sqrt{n}}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}\,t_1$$

<u>ОМП</u>. $\tilde{\theta}_2 = \frac{n+1}{2n+1}\,x_{max}$

$p(x,\theta) = \frac{1}{\theta}\mathbb{1}(\theta; 2\theta)$ — непр. дифф. при $\theta > 0$

$$I(\theta) = \int_{A(\theta)}\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2 p(x,\theta)\,dx = \int_{\theta}^{2\theta}\frac{1}{\theta^2}\cdot\frac{1}{\theta}\,dx =$$

$= \frac{1}{\theta^3} \cdot \theta = \frac{1}{\theta^2}$ непр. и >0

$\int_{\theta}^{2\theta} \frac{\partial p(x,\theta)}{\partial \theta} dx \pm \Rightarrow \int_{\theta}^{2\theta} -\frac{1}{\theta^2} \cdot dx = -\frac{1}{\theta^2} \quad \theta \neq 0$

$\Rightarrow$ не явл. регулярной $\Rightarrow$ нельзя применить асимпт. оценку по ОМП.

5 Т 6)

$$p(x) = \begin{cases} \frac{\theta-1}{x^{\theta}}, & x \geq 1; \\ 0, & x < 1 \end{cases} \qquad \theta > 1$$

$\vec{x}_n$ – выборка

а) $L(\theta) = \prod_{i=1}^{n} p(x,\theta) = \cancel{\frac{\theta-1}{x^{\theta}} I(x_i \geq 1)}$

$= \prod_{i=1}^{n} \frac{\theta-1}{x_i^{\theta}} = (\theta-1)^n \left(\prod_{i=1}^{n} x_i\right)^{-\theta} =$

$\ln L = n \ln(\theta-1) - \theta \ln \prod x_i$

$(\ln L)' = \frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i = 0 \qquad \frac{\partial^2 \ln L}{\partial \theta^2} = -\frac{n}{(\theta-1)^2} < 0 \Rightarrow$ макс.

$\tilde{\theta} - 1 = \frac{n}{\sum \ln x_i}; \quad \underbrace{\tilde{\theta} = 1 + \frac{n}{\sum \ln x_i}}$ ОМП

в) $p(x,\theta)$ – непр. дифф. по θ, для $\theta > 1$

$A(\theta) \int \frac{\partial p(x,\theta)}{\partial \theta} dx = \int_1^{+\infty} \left(\frac{1}{x^{\theta}} - \frac{\theta^2}{x^{\theta+1}} + \frac{1}{x^{\theta+1}}\right) dx =$

$= \left(\frac{-1}{x^{\theta-1}(\theta-1)} + \frac{\theta^2}{x^{\theta+2}(\theta+2)} - \frac{1}{x^{\theta+1}(\theta+2)}\right)\Big|_1^{+\infty} = \frac{1}{\theta-1} - \frac{\theta^2}{\theta+2} + \frac{1}{\theta+2} =$

и. рег

$\frac{\alpha}{2}\cdot 5000$ $\quad \beta\cdot 5000 \quad$ $\frac{\alpha}{2}\cdot 5000$

$\frac{1}{\theta-2} + \frac{1-\theta^2}{\theta+2}$

в) $p(x,\theta)$ – непр. дифф. по θ на $\theta>1$

$$\int_1^{+\infty} \frac{\partial p(x,\theta)}{\partial\theta}\,dx = \int_1^{+\infty}\left(\frac{1}{x^\theta} - \theta\cdot\frac{1}{x^\theta}\cdot\ln x + \frac{1}{x^\theta}\ln x\right)dx =$$

$$= \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} + \int_1^{+\infty}\ln x\,dx^{1-\theta} = \frac{-1}{1-\theta} +$$

$$+ \ln x\cdot x^{1-\theta}\Big|_1^{+\infty} - \int_1^{+\infty} x^{1-\theta}\frac{1}{x}\,dx = \frac{-1}{1-\theta} - \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} = 0$$

$$I(\theta) = \int_1^{+\infty}\left(\frac{\partial\ln(p(x;\theta))}{\partial\theta}\right)^2 p(x,\theta)\,dx =$$

$$\ln p = \ln(\theta-1) - \theta\ln x$$

$$\frac{\partial\ln p}{\partial\theta} = \frac{1}{\theta-1} - \ln x$$

$$= \int_1^{+\infty}\left(\frac{1}{\theta-1} - \ln x\right)^2\cdot\frac{\theta-1}{x^\theta}\,dx = \int_1^{+\infty}\left(\frac{1}{x^\theta(\theta-1)} - 2\ln x\cdot\frac{1}{x^\theta} +\right.$$

$$\left. + \frac{\theta-1}{x^\theta}\ln^2 x\right)dx = \frac{-x^{1-\theta}}{(\theta-1)^2}\Big|_1^{+\infty} - \int_1^{+\infty}\frac{2\ln x}{1-\theta}\,dx^{1-\theta} =$$

$$= \int_1^{+\infty}\ln^2 x\,dx^{1-\theta} = \frac{+1}{(\theta-1)^2} - \frac{2}{1-\theta}\left(\ln x\cdot x^{1-\theta}\Big|_1^{+\infty} - \int_1^{+\infty} x^{1-\theta}\cdot\frac{1}{x}\,dx\right) +$$

$$+ \left(\ln^2 x\cdot x^{1-\theta}\Big|_1^{+\infty} - \int_1^{+\infty} x^{1-\theta}\cdot 2\ln x\cdot\frac{1}{x}\,dx\right) = \frac{+1}{(\theta-1)^2} + \frac{2}{1-\theta}\left(\frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty}\right) =$$

$$+2\int_1^{+\infty} \ln x \cdot x^{-\theta}\,dx = \frac{+1}{(\theta-1)^2} + \frac{2}{(1-\theta)^2} + \frac{2}{1-\theta}\left(x^{1-\theta}\ln x\Big|_1^{+\infty} - \int_1^{+\infty} x^{1-\theta}\cdot\frac{1}{x}\,dx\right)$$

$$= \frac{+1}{(\theta-1)^2} + \frac{2}{(1-\theta)^2} + 2\cdot\frac{x^{1-\theta}}{(1-\theta)^2}\Big|_1^{+\infty} = \frac{+1}{(\theta-1)^2} + \frac{2}{(1-\theta)^2} + \frac{2}{(\theta-1)^2}$$

непр. на Θ; $I(\theta) > 0$

$\Rightarrow$ модель регулярна

$$\int_1^{+\infty} \frac{\partial^2 p(x,\theta)}{\partial\theta^2}\,dx = \int_1^{+\infty}\left(\frac{-1}{x^\theta}\ln x - \ln x\cdot\frac{1}{x^\theta} + \theta\cdot\frac{1}{x^\theta}\cdot\ln^2 x - \right.$$

$$\left. - \frac{1}{x^\theta}\ln^2 x\right)dx = \int_1^{+\infty}\frac{2\ln x}{\theta-1}\,dx^{1-\theta} = \int_1^{+\infty}\ln^2 x\cdot$$

$$\cdot\, d\left(x^{1-\theta}\right) = \frac{2}{\theta-1}\left(\int_1^{+\infty}\frac{x^{1-\theta}}{x}\,dx - \int_1^{+\infty} x^{1-\theta}\cdot 2\ln x\cdot\frac{1}{x}\,dx\right) =$$

$$= \frac{2}{\theta-1}\cdot\frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} + 2\,\frac{1}{1-\theta}\int_1^{+\infty}\ln x\,dx^{1-\theta} =$$

$$= \frac{2}{(\theta-1)^2} + 2\,\frac{1}{1-\theta}\int_1^{+\infty}\frac{x^{1-\theta}}{x}\,dx = \frac{2}{(\theta-1)^2} + 2\,\frac{1}{1-\theta}\cdot\frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} =$$

$$= \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0 \Rightarrow \text{модель сильно}$$

регулярна

B – открытое; p дважды непр. дифф.

$\Rightarrow$ вып. теорема для ОМП.

$\frac{\alpha}{2}\cdot 5000$ $\quad \beta\cdot 5000$ $\quad \frac{\alpha}{2}\cdot 5000$

$$F(x_p) = p = 0{,}5$$

$$F(x) = \int_1^x \frac{\theta-1}{t^{\theta}}\,dt = \frac{t^{-\theta+1}(\theta-1)}{(1-\theta)}\Big|_1^x =$$

$$= -\left(\frac{1}{x^{\theta-1}} - 1\right) = \frac{-1}{x^{\theta-1}} + 1 = \frac{1}{2}$$

$$x_{0,5}^{\theta-1} = 2; \quad x_{0,5} = 2^{\frac{1}{\theta-1}}; \quad f(\theta) = 2^{\frac{1}{\theta-1}}$$

$$\nabla f(\theta) = 2^{\frac{1}{\theta-1}} \cdot \ln 2 \cdot \left(-\frac{1}{(\theta-1)^2}\right)$$

$$I(\theta) = \frac{1}{(\theta-1)^2}; \quad I^{-1}(\theta) = (\theta-1)^2$$

$$\sigma^2(\theta) = \nabla^T f(\theta)\, I^{-1}(\theta)\, \nabla f(\theta)$$

$$\sigma^2(\theta) = \left(2^{\frac{1}{\theta-1}} \ln 2 \cdot \left(-\frac{1}{(\theta-1)^2}\right)\right)^2 \cdot (\theta-1)^2$$

$$\sqrt{n}\,\frac{g(\tilde\theta) - g(\theta)}{\sigma(\theta)} \rightsquigarrow N(0;1)$$

$$\sqrt{n}\,\frac{2^{\frac{1}{\tilde\theta-1}} - 2^{\frac{1}{\theta-1}}}{2^{\frac{1}{\theta-1}} \ln 2 \cdot \left(+\frac{1}{(\tilde\theta-1)^2}\right)(\tilde\theta-1)} \rightsquigarrow N(0;1)$$

$$t_1 = u_{\frac{1-\beta}{2}}; \quad t_2 = u_{\frac{1+\beta}{2}}$$

$$P\left(t_1 < \frac{\sqrt{n}(\tilde\theta-1)\left(2^{\frac{1}{\tilde\theta-1}} - 2^{\frac{1}{\theta-1}}\right)}{2^{\frac{1}{\theta-1}} \ln 2} < t_2\right) = \beta$$

$$\Rightarrow \frac{2^{\frac{1}{\tilde\theta-1}} \ln 2}{\sqrt{n}(\tilde\theta-1)}\, t_1 < 2^{\frac{1}{\tilde\theta-1}} - x_{0,5} < t_2 \frac{2^{\frac{1}{\tilde\theta-1}} \ln 2}{\sqrt{n}(\tilde\theta-1)}$$

$$\underbrace{2^{\frac{1}{\tilde\theta-1}} - \frac{2^{\frac{1}{\tilde\theta-1}}\ln 2}{\sqrt{n}(\tilde\theta-1)}\, t_2 < x_{0,5} < 2^{\frac{1}{\tilde\theta-1}} - \frac{2^{\frac{1}{\tilde\theta-1}}\ln 2}{\sqrt{n}(\tilde\theta-1)}\, t_1}$$

c) $\sqrt{n}\,\dfrac{\tilde\theta-\theta}{\sigma(\tilde\theta)} \rightsquigarrow N(0;1)$

$g(\theta)=\theta;\quad \nabla g(\theta)=1$

$\sigma^2(\theta)=1\cdot(\theta-1)^2\cdot 1$

$\sqrt{n}\,\dfrac{\hat\theta-\theta}{\tilde\theta-1} \rightsquigarrow N(0;1)$

$t_1 = u_{\frac{1-\beta}{2}};\quad t_2 = u_{\frac{1+\beta}{2}}$

$\sqrt{n}\,\dfrac{\tilde\theta - \theta}{\frac{n}{\sum \ln x_i}} \rightsquigarrow N(0;1)$

$P\left(t_1 < \frac{\sqrt{n}}{n}(\tilde\theta-\theta)\sum\ln x_i < t_2\right)=\beta$

$\cancel{\tilde\theta - \frac{\sqrt{n}}{n} t_2 \sum \ln x_i < \theta < \tilde\theta - \frac{\sqrt{n}}{n} t_1 \sum \ln x_i}$

$$\underbrace{\tilde\theta - \frac{\sqrt{n}}{\sum\ln x_i}\, t_2 < \theta < \hat\theta - \frac{\sqrt{n}}{\sum\ln x_i}\, t_1}$$